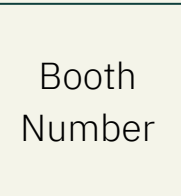


Aoun: A Low-Cost, Two-Way LoRa Distress and Tracking System that Connects Lost Persons to Rescue Teams Beyond Cellular Coverage



Student Name: Dana Mohammed Alanazi (Team Leader), Renad Metab Aldosari, Jawaher Faisal Alsharif, Hailah Abdulrahman Alhejjei, Abrar Hassan Alqarni

ID:

Introduction

People lost in deserts, mountains and remote areas are often far from any cellular coverage, so rescue teams must search wide areas without an exact location.

Phones need coverage. Satellite messengers need subscriptions. Beacons alert a distant centre and give the sender little or no confirmation. A lost person who does not know help is coming may panic and move away from the search area.

Aoun closes this gap with a low-cost, infrastructure-free, **two-way** radio link.

Objectives

1. Send the GPS location with one SOS press, with no network needed
2. Return instant confirmation to the lost person (green LED + message)
3. Show the live location on the rescue team's screen

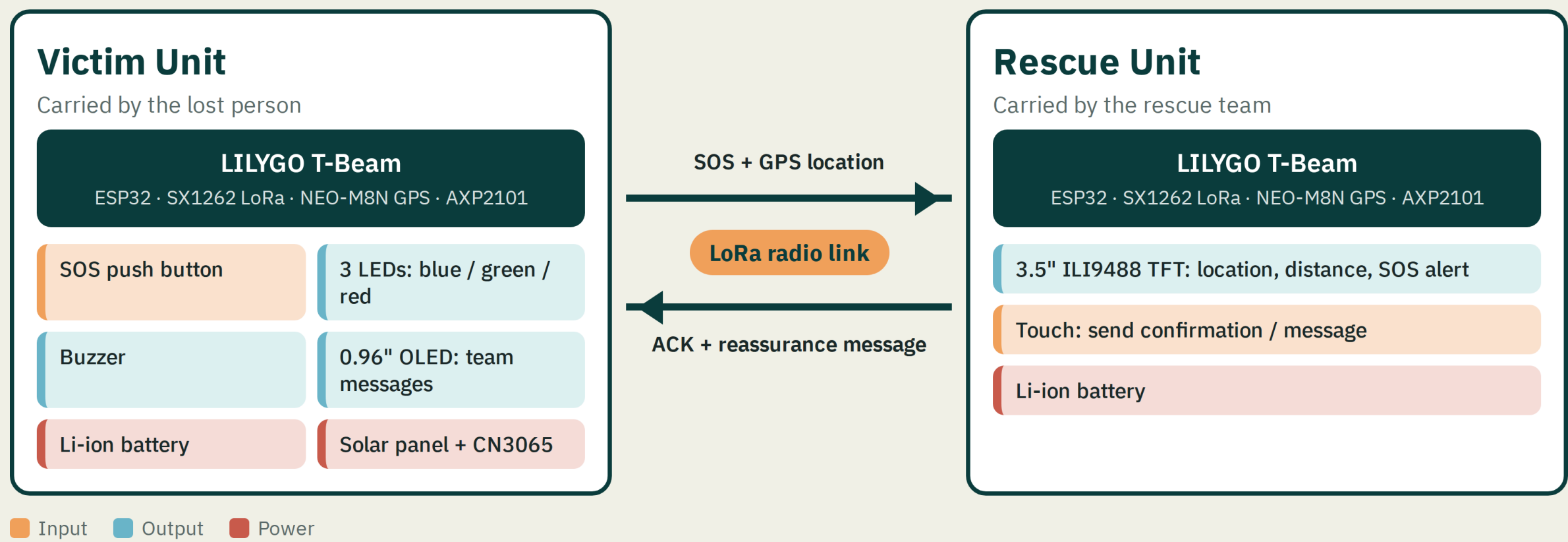


Fig. 1: Aoun system block diagram. Both units use the same board; no cellular network or satellite is needed.

Methodology

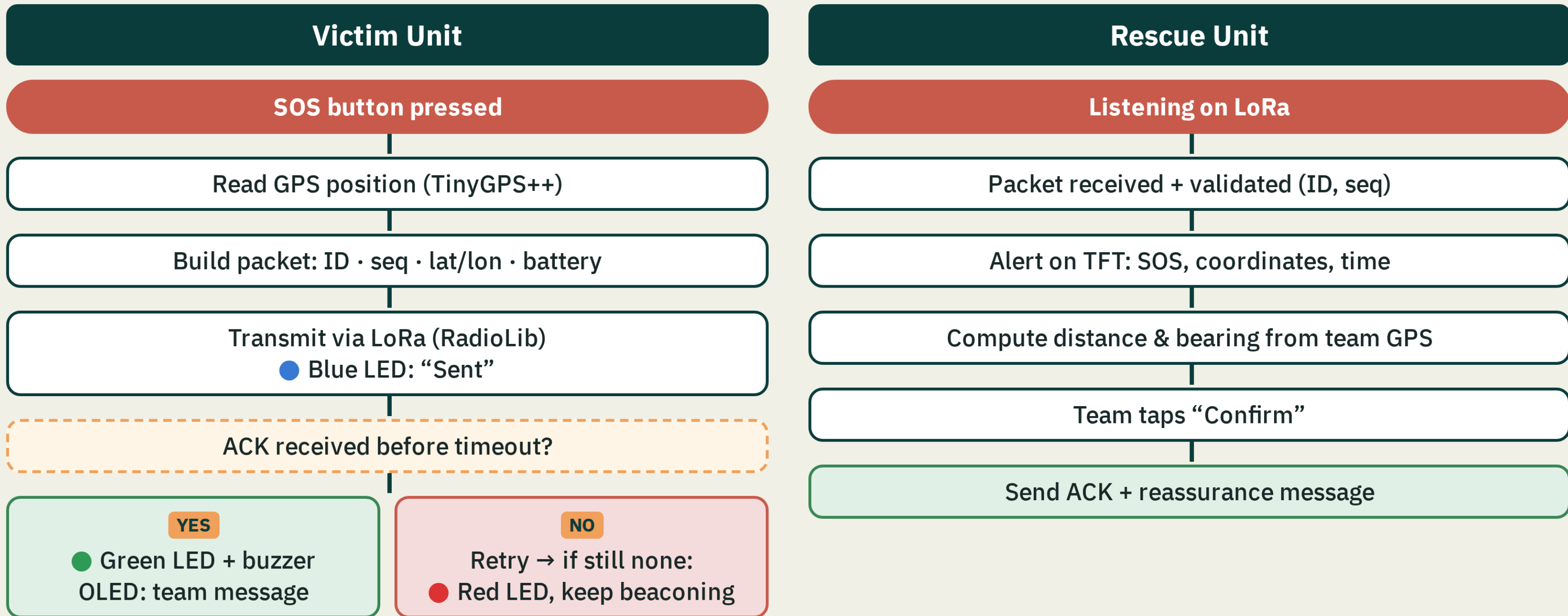


Fig. 2: Operating flow. The two-way confirmation loop is Aoun's core innovation.

Hardware: 2 × LILYGO T-Beam (ESP32 + SX1262 LoRa + NEO-M8N GPS + AXP2101). Victim unit: SOS button, 3 LEDs, buzzer, OLED, solar charging. Rescue unit: 3.5" TFT touch display.

Firmware: Arduino C++ · RadioLib · TinyGPS++ · TFT_eSPI · Adafruit SSD1306.

Security: AES-128-GCM authenticated encryption · sequence numbers against replay · registered-device whitelist. **Why**

LoRa: long range, low power, licence-free band, no towers.

Results

10 / 10

security tests passed

5 / 5

attack types rejected

37 bytes

encrypted SOS packet

62 bytes

ACK + reassurance message

≈ 0.34 s

SOS airtime at SF9 (2.5 s at SF12)

0

location bytes readable over the air

Security test case	Result	Field validation plan (after assembly)	
		Metric	Method
Valid SOS decrypts correctly	PASS	Range (line of sight)	0.5 / 1 / 2 / 5 km
Location hidden in radio packet	PASS	Delivery rate	50 SOS per distance
Replayed packet rejected	PASS	Confirmation time	Timestamped logs
Flipped bit in ciphertext rejected	PASS	Battery (+ solar)	Continuous beaconing
Modified header rejected	PASS	On-device crypto	ESP32 timing test
Wrong-key packet rejected	PASS	Airtime was calculated with the Semtech LoRa time-on-air formula (BW 125 kHz, CR 4/7). Encryption adds only 23 bytes per packet. Stage: Proof of Concept. Hardware assembly is in progress.	
Unregistered device rejected	PASS		
ACK matched to last SOS	PASS		
Accepted after victim reboot	PASS		
Truncated packet rejected	PASS		

Table 2: Security verification. aoun_crypto.h compiled against Mbed TLS 3.6.2 on a PC host.

Innovation

Two-way reassurance, not just an alarm.

Aoun is the only option compared here that tells the lost person, instantly and visibly, that the nearby team received the call. They stay calm and stay in place, which makes them easier to find. It needs no network, subscription or satellite.

Feature	Mobile phone	Satellite messenger	Locator beacon (PLB)	Aoun
Works without cellular	✗*	✓	✓	✓
Confirmation to lost person	✗	✓	Limited	✓ LED + message
Alert goes directly to nearby team	✗	✗	✗	✓
No subscription	✗	✗	✓	✓
Low cost	—	✗	✗	✓

* Some recent phones offer satellite SOS on selected models and in selected countries.

Table 1: Aoun compared with existing emergency options.

Conclusion

Aoun sends a lost person's GPS location to a rescue team over a direct LoRa link, with no cellular network, and returns a confirmation that reassures them. Every message is encrypted and authenticated. The security layer passed all 10 verification tests, and an SOS fits in a 37-byte packet.

Impact: faster searches, calmer victims and lower rescue costs. It can serve Civil Defense, Border Guard, volunteer rescue teams, desert trips and large events such as Hajj.

Readiness: Proof of Concept. The design, firmware and security layer are complete and verified. Hardware assembly and field testing are in progress.

Future Work & References

1. Complete field validation: range, delivery rate, confirmation time, battery
2. Mesh relaying between units to cover large search areas
3. Rugged waterproof enclosure + automatic fall / no-motion SOS
4. Multi-team dashboard and a pilot with a rescue team

References: scan the QR code (top right) for the project page, security report and references.